

Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur
FOUR YEAR B. TECH. COURSE
(Revised Curriculum as per AICTE Model Curriculum)
Scheme for Computer Engineering

Fifth Semester:-

S. N.	Subject Code	Subject	Teaching Scheme			Evaluation Scheme			Credits	Category
			L	T	P	CA	UE	Total		
1	BTCME501T	Database Management System	3	1	-	30	70	100	4	PCC-CS
2	BTCME501P	Database Management System – Lab	-	-	2	25	25	50	1	PCC-CS
3	BTCME502T	Computer Graphics	3	-	-	30	70	100	3	PCC-CS
4	BTCME502P	Computer Graphics –Lab	-	-	2	25	25	50	1	PCC-CS
5	BTCME503T	Java Programming	3	-	-	30	70	100	3	PCC-CS
6	BTCME503P	Java Programming - Lab	-	-	2	25	25	50	1	PCC-CS
7	BTCME504T	Elective-I	3	-	-	30	70	100	3	PEC-CS
8	BTCME505T	Humanities - II – Effective Technical Communication	2	1	-	15	35	50	3	HSMC
9	BTCME506T	Yoga & Meditation	-	-	2	-	-	-	Audit	MC
Total			16	02	06	210	390	600	19	

Elective-I:

BTCME504.1T - Microcontrollers & Applications,

BTCME504.2T - Artificial Intelligence,

BTCME504.3T - Software Engineering

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RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FOUR YEAR BACHELOR OF ENGINEERING (B.TECH.) DEGREE COURSE

SEMESTER: V (C.B.C.S.)

BRANCH: COMPUTER ENGINEERING

Subject : **DATABASE MANAGEMENT SYSTEM**

Subject Code : **BTECHCME501T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
(3 Hrs. +1Hr.) (L+ T)	4	30	70	100

Aim: To realize the concepts of database Management System

Prerequisite(s): File structure, set theory concepts from discrete mathematics, Object Oriented Concepts

Course Objectives: (02-04)

1	To provide students with basic concepts in databases both in terms of usage and Implementation
2	To make the students understand all requirement and operations that the analyst needed to analyse, design, and implement the systems
3	To develop real time application using the Structure Query Language

Course Outcomes: (06)

At the end of this course Student are able to:

CO1	Explain the architecture and functioning of database management system as well as associated tools and techniques. Compare file processing system and DBMS
CO2	Illustrate database query and manipulation languages such as SQL (Structured query language) and Interpret real problem with SQL query
CO3	Classify various data models and construct Entity _Relationship diagram Apply normalization form to convert database tables into normal forms.
CO4	Query Processing and Query Optimization Measure of query cost for various Operators
CO5	Transaction Management Explain role of various components in the transaction management
CO6	Understand the concept of Concurrency Control system

Unit I: Introduction to DBMS**[6 Hours]**

General introduction to database systems, Database - DBMS distinction, Approaches to building a database, Data models, Three-schema architecture of a database, Challenges in building a DBMS, Various components of a DBMS

Unit II: Query Language**[8 Hours]**

Relational Data Model, Concept of relations, Schema-instance distinction, Keys, referential integrity and foreign keys, Relational algebra operators, Tuple relation calculus, Domain relational calculus, SQL, PL/SQL .

Unit III: Data Models and Relational Database Design**[8 Hours]**

Evolution of Data Models, Entity Relationship Model, Development of ER Diagrams, Extended Entity Relationship Model. Relational model: Logical View of Data, Concepts of Functional dependency, Normalization 1NF, 2NF, 3NF, BCNF.

Unit IV: Query Processing & optimization**[6 Hours]**

Overview: Query Processing and Optimization, measures of query cost estimation in query Optimization, pipelining and Materialization, Structure of query evaluation plans.

Unit V: Transaction management**[8 Hours]**

Transaction concepts, properties of transactions, serializability of transactions, testing for serializability, transaction isolation levels and implementations. **Concurrency control:** Lock Based Protocol, Deadlock Handling, Time-stamp Based Protocols, and Validation Based Protocols.

Text Books :-

1. Database System Concepts by AviSilberschatz , Henry F. Korth , S. Sudarshan, Tata McGraw Hill, Fifth Edition
2. An Introduction to database management Systemby Bipin Desai

Reference Books:-

1. Introduction to Database Management Systems by Kahate



Subject : **DATABASE MANAGEMENT SYSTEM**

Subject Code : **BTCME501P**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
2 Hrs (Practical)	1	25	25	50

Practical List

Practical based on above syllabus (Oracle , SQL, post gress Sql , Sqlite etc). Teachers can emphasis to do mini application using database knowledge to students. Faculty can assign case study to the students.

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Subject : Computer Graphics

Subject Code : BTCME502T

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To understand the concept of designing the computer graphics

Prerequisite(s): None

Course Objective

1	To introduce students with fundamental concepts and theory of Computer Graphics.
2	To understand & exhibit the line, circle, ellipse, polygon filling, clipping, hidden surface removal, Curves & surface, rendering algorithms and 2D & 3D transformations
3	To allows students to develop programming skills in Computer Graphics.

Course Outcomes:

Upon the completion of the course students will be able to :-

CO1	Understand the basics of computer graphics, and different graphics systems. Apply and compare the algorithms for drawing 2D images also explain aliasing & anti-aliasing
CO2	Apply and compare the Polygon Filling algorithms.
CO3	Solve the problems on viewing transformations. Analyze and apply clipping algorithms.
CO4	Analyze and apply transformation on 2D & Explain the projection 3D Graphics.
CO5	Explain the hidden surface removal, Curves & surface rendering algorithms

UNIT I: Introduction to Computer Graphics

[08 Hours]

Geometry and line generation: points, lines, planes, pixels and frames buffers, types of display devices and its architecture DDA and Bresenham's algorithms for line generation, Bresenham's algorithm for circle generation, aliasing, anti-aliasing and its techniques.

UNIT II:

[07 Hours]

Graphics primitives, Display files, Segment Table, Operations on Segment, data structures for

segments.

Polygon generation, Polygon filling algorithms: Simple order Edged List, Edge Fill, Edge Flag, Seed Fill, Scan Line Seed Fill

UNIT III:

[07 Hours]

Windowing and clipping: window, viewport, viewing transformations, NDC (normalized device co-ordinates), WCS (World co-ordinates System)

Clipping, line : Cohen-Sutherland, Cyrus-Beck, Midpoint Subdivision algorithms & Polygon clipping: Sutherland-Hodgeman

UNIT IV:

[07 Hours]

2D transformations: scaling, rotation, translation, rotation about arbitrary point, reflections, shearing.

3D Graphics: 3D Transformation, parallel, perspective and isometric projections.

UNIT V:

[07 Hours]

Hidden surfaces and line removal: Painter's, Z-buffer, Warnock's, Back-face Removal algorithm

Curves and surfaces: Methods of interpolation, Bezier and B-splines, surface rendering methods: Gouraud Shading, Phong Shading, Constant Intensity Shading, Fast Shading.

Text Books:-

1. Procedural elements for computer graphics by David F. Rogers, Mc-Graw Hill.
2. Computer Graphics 'C' Version, Second Edition By Donald Hearn and M. Pauline Baker, Pearson publication
3. Mathematical elements for computer graphics by David Rogers and J. Alan Adams, Tata Mcgraw Hill Education Private Limited
4. Computer graphics principles and practice in C by Foley, Vandam, Feiner and Huges (Pearson)
5. Principles of interactive computer graphics by Newman and Sproul.





Subject : **Computer Graphics**

Subject Code : **BTCME502P**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
2 Hrs (Practical)	1	25	25	50

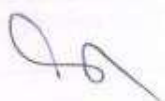
Practical List

Practical based on above syllabus.











Subject: Java Programming

Subject Code: BTCME503T

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To inculcate the Java programming concepts among the students in order to strengthen the programming logic.

Prerequisite(s): Basic programming Languages like C, C++ etc.

Course Objectives:

1	To teach principles of object oriented programming paradigm including abstraction, encapsulation, inheritance and polymorphism.
2	Impart fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc.
3	Inculcate concepts of inheritance to create new classes from existing one & Design the classes needed given a problem specification;
4	Familiarize the concepts of packages and interfaces.
5	Facilitate students in handling exceptions.

Course Outcomes:

CO1	Analyze the necessity for Object Oriented Programming paradigm over structured programming and become familiar with the fundamental concepts in OOP like encapsulation, Inheritance and Polymorphism
CO2	Design and develop java programs, analyze, and interpret object oriented data and report results.
CO3	Design an object oriented system, AWT components and multithreaded processes as per needs and specifications.
CO4	It help to succeed in competitive examinations like GATE, Engineering services, Technical interviews etc.
CO5	Plan their career in java based technologies like HADOOP etc

Unit I:

[7 Hours]

JAVA BASICS: Review of Object oriented concepts, History of Java, JVM architecture, Data types, Variables, Scope and life time of variables, arrays, operators, control statements, simple java program, Static block, Static Data, Static Method, String handling and String Buffer Classes.

Unit II:

[7 Hours]

OBJECT & CLASS: Object Oriented Programming , Class Fundamentals, Object & Object reference, Object Life time & Garbage Collection, Creating and Operating Objects, Constructor & initialization code block, Access Control, Modifiers, methods, Inner Class & Anonymous Classes.

Unit III:**[8 Hours]**

INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and super keyword, Method Overloading, Method overriding, Abstract classes, Dynamic method dispatch, Usage of final keyword.

PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces.

Unit IV:**[8 Hours]**

INPUT AND OUTPUT AND FILE HANDLING: Concepts of streams, Stream classes- Byte and Character stream, reading console Input and Writing Console output, Wrapper Classes, File Handling.

EXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes.

Unit V:**[6 Hours]**

MULTITHREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication.

Text Books:

1. Herbert schildt (2010), The complete reference, 7th edition, Tata Mc graw Hill, New Delhi

Reference Books:

1. Head First Java, O'rielly publications
2. T. Budd (2009), An Introduction to Object Oriented Programming, 3rd edition, Pearson Education, India.
3. J. Nino, F. A. Hosch (2002), An Introduction to programming and OO design using Java, John Wiley & sons, New Jersey.
4. Y. Daniel Liang (2010), Introduction to Java programming, 7th edition, Pearson education, India



Subject : **Java Programming**

Subject Code : **BTCME503P**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
2 Hrs (Practical)	1	25	25	50

Practical List

Practical based on above syllabus.

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Subject: **Microcontrollers & Applications (Elective-I)**

Subject Code: **BTCME504.IT**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To realize the concepts of Microcontroller

Prerequisite(s): Microprocessor, Instructions set

Course Objectives: (02-04)

1	To provide understanding of basics of microcontroller
2	To enable students to understand features of 8051 microcontroller
3	Knowledge of working of microcontroller systems in various fields.

Course Outcomes: (06)

At the end of this course Student are able to:

CO1	Understand the difference between microprocessor & microcontrollers
CO2	Analyze the architecture of 8051 microcontroller.
CO3	Develop programs in assembly language.
CO4	Interface peripherals to microcontroller.
CO5	To impart the basic concepts of serial communication in 8051
CO6	Illustrate the use of microcontroller in different applications.

UNIT I:

[6 Hours]

Introduction to Microcontroller: Block diagram of microcontroller: Comparison between microprocessor and microcontroller, Embedded Systems, Selection factors of microcontroller (Architecture type, speed, Word size, instruction set, memory, and I/O capability)

Types of Buses-Address, Data, Control, common features of Microcontrollers: On-chip Oscillator, program and data memory, I/O Ports, Watchdog- timer reset, Timers, Counters, and Interrupts.



UNIT II :**[8 Hours]**

The Microcontroller 8051: Features of 8051 microcontroller. Block diagram of 8051- program status word (PSW), accumulator, and program counter. Memory organization – RAM & ROM, register banks and stack, Pin out diagram- description of pins, special function registers (SFRs), I/O port organization, interrupts.

UNIT III:**[8 Hours]**

8051 Instruction Set: Addressing Modes, Data Transfer instructions, Arithmetic instructions, Logical instructions, Branch instructions, and Bit manipulation instructions. Simple Assembly language program examples (without loops) to use these instructions.

UNIT IV:**[8 Hours]**

8051 Timers and Serial Port: 8051 Timers and Counters – Operation and Assembly language programming to generate a pulse using Mode-1 and a square wave using Mode- 2 on a port pin. 8051 Serial Communication- Basics of Serial Data Communication, RS- 232 standard, 9 pin RS232 signals, Simple Serial Port programming in Assembly and C to transmit a message and to receive data serially.

UNIT V :**[6 Hours]**

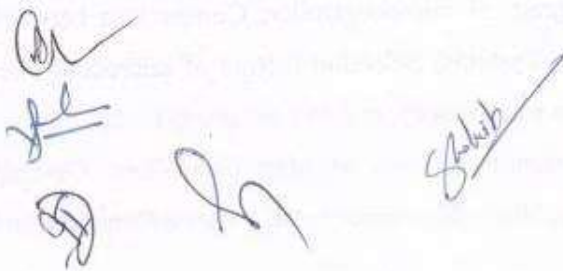
Applications based on IoT: Introduction of the Internet of Things, Types of sensors, Types of actuators, Introduction of Arduino Interfacing of the sensors and actuators with Arduino. Programming in Arduino. Design of a minor project based on Arduino

Text Books :-

1. Kenneth.J.Ayala. The 8051 microcontroller, 3rd edition, Cengage learning
2. The 8051 Microcontrollers: Architecture, Programming and Applications, Rao Dr. K Uma, Pearson Education India, New Delhi, (Latest edition)

Reference Books:-

1. Microcontroller theory & application. Ajay Deshmukh, Tata McGraw- Hill
2. Microcontroller Architecture, programming, interfacing, & system design, Raj Kamal, Pearson Edition.

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Subject: **Artificial Intelligence (Elective-I)**

Subject Code: **BTCME504.2T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To understand the concept of Artificial Intelligence basics.

Prerequisite(s): None

Course Objectives:- (02-04)

1	To understand the various characteristics of Intelligent agents.
2	To learn the different search strategies in AI.
3	To learn to represent knowledge in solving AI problems.
4	To understand the ways of planning and acting in the real world.

Course Outcomes:- (06)

At the end of this course Student are able to:

CO1	To design a plan for the real world problems and mapping it to the digital world
CO2	To identify problems that are amenably solved by AI methods
CO3	To apply the different knowledge representation for solving the problems
CO4	To apply the various planning strategies for real world problems.
CO5	To identify the uncertainty of solution to any problems and resolve it

UNIT I :- Introduction

[8 Hrs]

Introduction - Definition - Future of Artificial Intelligence - Characteristics of Intelligent Agents – Typical Intelligent Agents - Problem Solving Approach to Typical AI problems.

UNIT II:- Problem Solving Methods

[8 Hrs]

Problem solving Methods - Search Strategies - Uninformed - Informed - Heuristics - Local Search Algorithms and Optimization Problems - Searching with Partial Observations -Backtracking Search

- Performance of search algorithms.

UNIT III:- Knowledge Representation

[7 Hrs]

First Order Predicate Logic - Unification - Forward Chaining - Backward Chaining - Resolution – Knowledge Representation using First order Predicate logic - Reasoning Systems.

UNIT IV:- Planning

[6 Hrs]

Planning with state-space search - partial-order planning - planning graphs - planning and acting in the real world- Plan generation systems.

UNIT V:- Uncertain Knowledge and Reasoning

[7 Hrs]

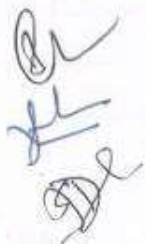
Uncertainty - review of probability - probabilistic Reasoning - Bayesian networks - inferences in Bayesian networks - Temporal models - Hidden Markov models.

Text Book:-

1. S. Russel, P. Norvig, "Artificial Intelligence – A Modern Approach", Third Edition, Pearson Education, 2015.

Reference Books:-

1. Kevin Night, Elaine Rich, Nair B., "Artificial Intelligence (SIE)", Third Edition, McGraw Hill, 2017.
2. Dan W. Patterson, "Introduction to AI and ES", Pearson Education, 2007



Subject: **Software Engineering (Elective – I)**

Subject Code: **BTCME504.3T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To Learn how to develop good quality software for a business system.

Prerequisite(s): None

Course Objectives: (02-04)

1	To comprehend the various software process models.
2	To understand the types of software requirements and SRS document.
3	To know the different software design and architectural styles.
4	To learn the software testing approaches and know about quality control and risk management.

Course Outcomes: (06)

At the end of this course Student are able to:

CO1	To compare and select a process model for a business system.
CO2	To identify and specify the requirements for the development of an application.
CO3	To develop and maintain efficient, reliable and cost effective software solutions.
CO4	To critically think and evaluate assumptions and arguments of the client.
CO5	To understand the different software testing.
CO6	To understand the risk management and quality of software.

Unit I:

[6 Hours]

Introduction to Software Engineering: The evolving role of software, Software myths.

Software engineering- A layered technology, a process framework.

Process models: The waterfall model, Incremental process models, Evolutionary process models, Agile Process model.

Unit II:

[8 Hours]

Software Requirements: Functional and non-functional requirements, User requirements, System requirements, software requirements document.

Requirements engineering process: Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management.

Requirements Analysis, Analysis Modeling Approaches, Data Modeling, Scenario-Based Modeling, Flow-Oriented Modeling, Class-based Modeling, Behavioral Model.

Unit III:

[9 Hours]

Design Engineering: Design process and Design quality, Design concepts, the design model.
Creating an architectural design: Software architecture, Data design, Architectural Design.

Object-Oriented Design: Objects and object classes, An Object-Oriented design process, Design evolution.

Performing User interface design: Golden rules, User interface analysis and design, interface analysis, interface design steps, Design evaluation.

Unit IV:

[6 Hours]

Testing Strategies: A strategic approach to software testing, test strategies for conventional software, Black- Box and White-Box testing, Validation testing, System testing, the art of Debugging.

Product metrics: Software Quality, Metrics for Analysis Model, Metrics for Design Model

Unit V:

[7 Hours]

Risk management: Reactive vs. Proactive Risk strategies, software risks, Risk identification, Risk projection

Quality Management: Quality concepts, Software quality assurance, Software Reviews, Formal technical reviews, Statistical Software quality Assurance, Software reliability.

Text books:

1. Software Engineering A practitioner's Approach, Roger S Pressman, 6th edition. McGrawHill International Edition.
2. Software Engineering, Ian Sommerville, 7th edition, Pearson education.

Reference books:

1. Software Engineering, A Precise Approach, Pankaj Jalote, Wiley India, 2010.
2. Software Engineering: A Primer, Waman S. Jawadekar, Tata McGraw-Hill, 2008



Subject: **Effective Technical Communication**

Subject Code: **BTCME505T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
(3 Hrs. +1Hr.) (L+ T)	3	15	35	50

Aim: To develop a confidence and communication skills among students in order to face the interviews confidently.

Prerequisite(s): None

Course Objectives: (02-04)

At the end of the semester,

1	Students will have enough confidence to face competitive examinations (IELTES/ TOEFL/CAT/ MAT/ XAT/SNAP/GMAT/GATE etc.)
2	They will also acquire language skills required to write their Reviews / Projects / Reports.
3	They will be able to organize their thoughts in English and hence face job interviews more confidently.

Course Outcomes: (06)

After completing the course, students will

CO1	Acquire knowledge of structure of language.
CO2	Be able to face competitive exams and the interview process and can become employable.
CO3	Develop business writing skills.
CO4	Become familiar with technology enabled communication and can develop technical and scientific writing skills.

Syllabus

Unit 1. Functional Grammar: [7Hrs]

Common errors, Transformation of Sentences- Change the Voice, Change the Narration, Simple, Compound Complex sentences, Use of Phrases, Idioms & Proverbs.

Unit II. English for Competitive Exams & Interview Techniques: [5 Hrs]

Word building, **English** words /phrases derived from other languages, Prefixes and Suffixes, Synonyms/Antonyms, Technical Jargons, Verbal Analogies, Give one word for, Types & Techniques of Interview.

Unit III. Formal Correspondence [5 Hrs]

Business Letters, (Enquiry, Quotation, Order, Complaint), Job applications and Resume Writing, e-mail etiquette, Writing Memorandum, Circulars, notices, Analytical comprehension

Unit IV. Technical & Scientific Writing:

[7 Hrs]

Features of Technical Writing, Technical Report writing (Accident, Feasibility, Trouble, Progress), Writing Scientific Projects, Writing Manuals, Writing Project Proposals, Writing Research papers.

Reference Books:

1. Effective technical Communication by Barun K. Mitra, Oxford University Press.
2. Technical Communication-Principles and Practice by Meenakshi Raman & Sharma, Oxford University Press, 2010,
3. Functional English for Technical Students by Dr. Pratibha Mahato and Dr. Dora Thompson, Himalaya Publishing House
4. How to Prepare a Research Proposal: Guidelines for Funding and Dissertations in the Social and Behavioral Sciences by Krathwohl & R David
5. Technical Writing- Process and Product by Sharon J. Gerson & Steven M. Gerson, 3rd edition, Pearson Education Asia, 2000
6. Developing Communication skills by Krishna Mohan & Meera Banerjee.

The block contains several handwritten marks. On the left, there are three vertically stacked initials: 'AK', 'SL', and 'JD'. To the right of these, there are two larger, more elaborate signatures. The first is a cursive signature that appears to be 'S. Mohan', and the second is a signature that appears to be 'Meera Banerjee'.